ITPM

IT Project Management

Unit 3 — Agile Project Management

Simple, Easy-to-Understand Notes

1. What is Agile?

Agile is a way of doing Project Management (PM) where the work is implemented step by step, rather than trying to do everything at once.

How an Agile Project Works

- The project is broken into small parts
- Each small part is assigned to a team
- The team works on that small project/part
- The team gets feedback from the customer
- Based on the feedback, the team plans changes
- The team improves the work continuously

Main Benefits of Agile

1. Work gets done faster
2. Changes can be applied easily
3. Customer feedback keeps the customer connected with the project
4. Problems can be found early
5. Team members work together

In short: Agile = doing the project in small steps → gathering feedback → improving the work → better project quality.

2. Agile Principles

These are the 12 core principles that guide how an Agile project is run.

#	Principle	Meaning
1	Customer Satisfaction	The most important goal — keeping the customer happy is the top priority in an Agile project
2	Welcome Changes	Accept changes at every stage of the project process, and implement them
3	Deliver Work Regularly	Give results to the customer step-by-step, on a regular basis
4	Work Together	Developers, testers, and analysts work together as one group
5	Trust the Team	Give the team freedom to do their work
6	Good Communication	Keep clear and open communication among the team
7	Working Product is Important	A working product matters more than heavy documentation
8	Keep a Steady Pace	Work at a normal, sustainable speed — not overloaded
9	Good Quality	Focus on maintaining high quality of work
10	Keep Things Simple	Avoid unnecessary complexity
11	Self-Organizing Team	The team decides and organizes its own work
12	Continuous Improvement	Keep improving the process and product regularly

3. Agile Methodology — General Idea

Agile methodology means developing the product in small parts, checking each part, getting feedback, and then making improvements.

Main Features

6. Small and frequent releases
7. Customer involvement
8. Easy to handle changes
9. Teamwork
10. Continuous improvement

Example: Restaurant Food Ordering App

If we build this app the Agile way, we break it into small parts such as:

- 1) Login & Registration
- 2) Restaurant Search
- 3) Food Ordering
- 4) Online Payment
- 5) Order Tracking

Each part is developed, tested, and reviewed separately instead of building the entire app in one go.

4. Scrum

Scrum is a popular Agile methodology where work is divided into fixed time periods called Sprints (usually 1–4 weeks each).

Example: If total work takes 3 months, it can be divided into Sprint 1, Sprint 2, Sprint 3, Sprint 4 — each sprint delivering a working part of the product.

4.1 Scrum Roles (Total 4 Roles)

#	Role	Responsibility
1	Product Owner	Represents the customer; decides the features needed in the product
2	Scrum Master	Acts as the team lead; removes problems/obstacles and supports the team (like a manager)
3	Developers	Design, develop, test, and make changes to the product
4	Scrum Team	The whole group (Product Owner + Scrum Master + Developers) working together

4.2 Scrum Events

#	Event	Meaning
1	Sprint Planning	Weekly/sprint-start planning of what work will be done
2	Daily Scrum	A short daily meeting to check work progress properly
3	Sprint Review	Review of the completed sprint work with stakeholders
4	Sprint Retrospective	Discussion on what went well and what can be improved for the next sprint

5. Kanban

Kanban is a simple Agile method that uses a virtual board to visually show the work progress.

Kanban Board Example

TO DO	In Progress	Done
Login	Payment	Registration
Search	Order Track	Profile

Tasks move from left to right across the board (To Do → In Progress → Done) as work progresses.

6. Extreme Programming (XP)

XP stands for Extreme Programming. It focuses mainly on:

- Software quality
- Good programming practices

Key Features of XP

11. Pair Programming — two developers work together on the same code
12. Test Driven Development — tests are written before writing the actual code
13. Continuous Feedback — regularly getting feedback to improve the product
14. Frequent Releases — releasing small updates often

7. Lean

Lean is an Agile approach focused on creating maximum customer value while reducing waste.

Types of Waste in Lean

15. Unnecessary features
16. Unnecessary meetings
17. Waiting time
18. Repeated work

Quick Revision Summary

Topic	One-line Takeaway
Agile	Step-by-step implementation of a project in small parts, with continuous customer feedback
Agile Principles	12 principles — customer satisfaction, welcoming change, regular delivery, teamwork, trust, communication, working product, steady pace, quality, simplicity, self-organizing team, continuous improvement
Agile Methodology	Develop in small parts → check each part → get feedback → improve
Scrum	Work divided into Sprints (1–4 weeks); 4 roles: Product Owner, Scrum Master, Developers, Scrum Team; 4 events: Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective

Kanban	Visual board method with columns: To Do, In Progress, Done
Extreme Programming (XP)	Focuses on software quality and good coding practices — Pair Programming, Test Driven Development, Continuous Feedback, Frequent Releases
Lean	Maximizes customer value while reducing waste (unnecessary features, meetings, waiting time, repeated work)